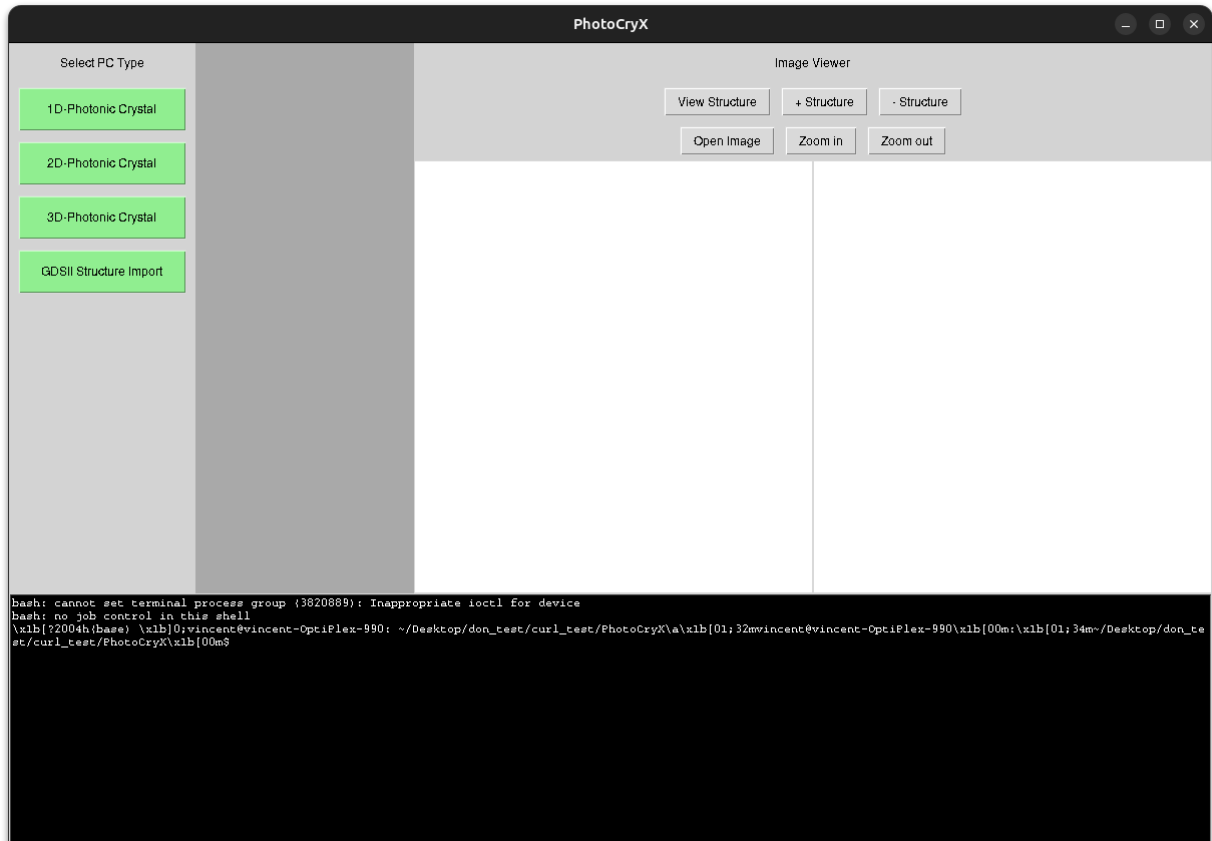


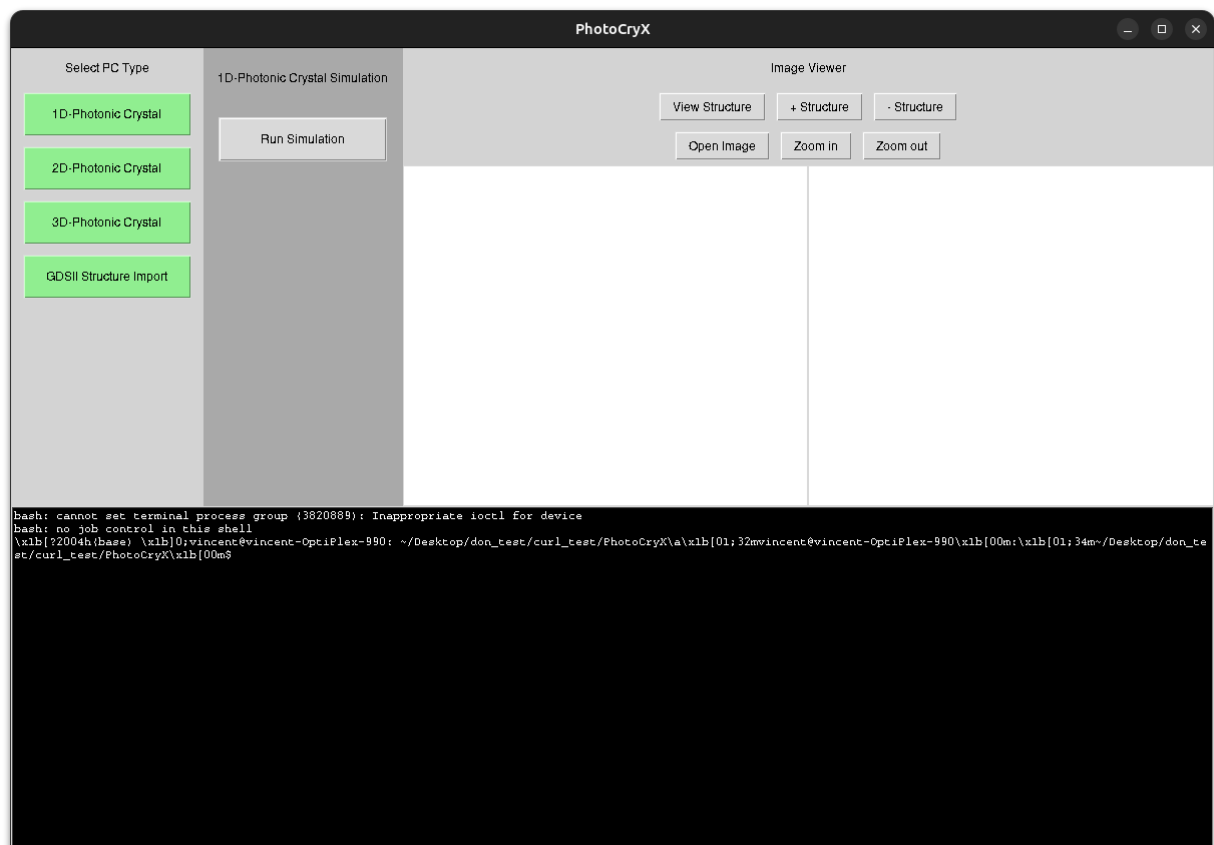
PhotoCryX tutorial using GUI

The initial interface of PhotoCryX is;



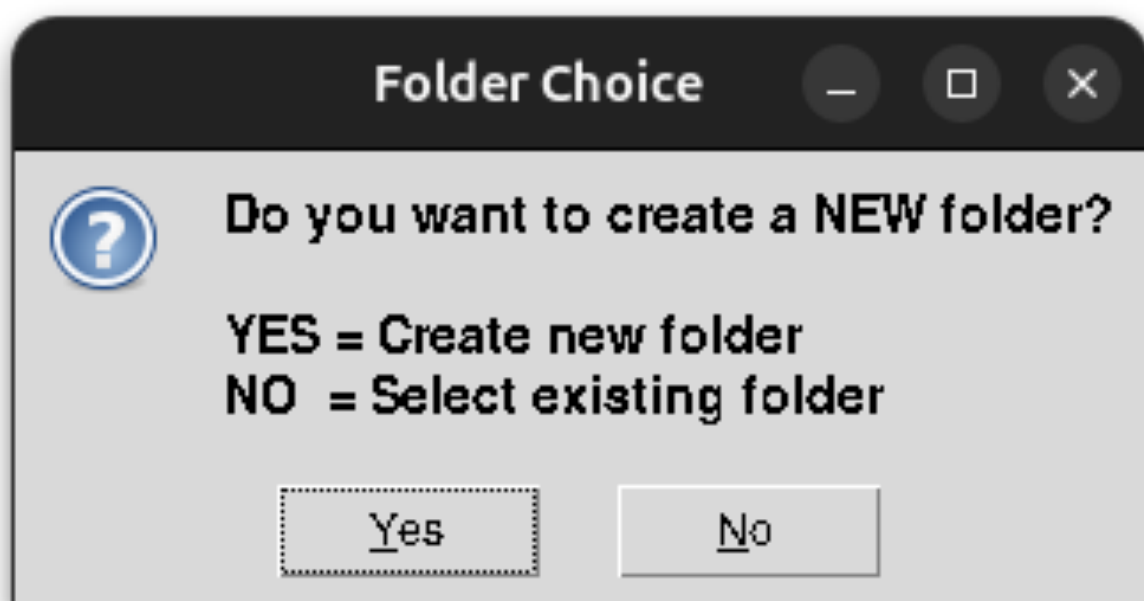
One-dimensional Photonic Crystal (1DPC)

The 1D interface has just the "Run Simulation" button as shown below

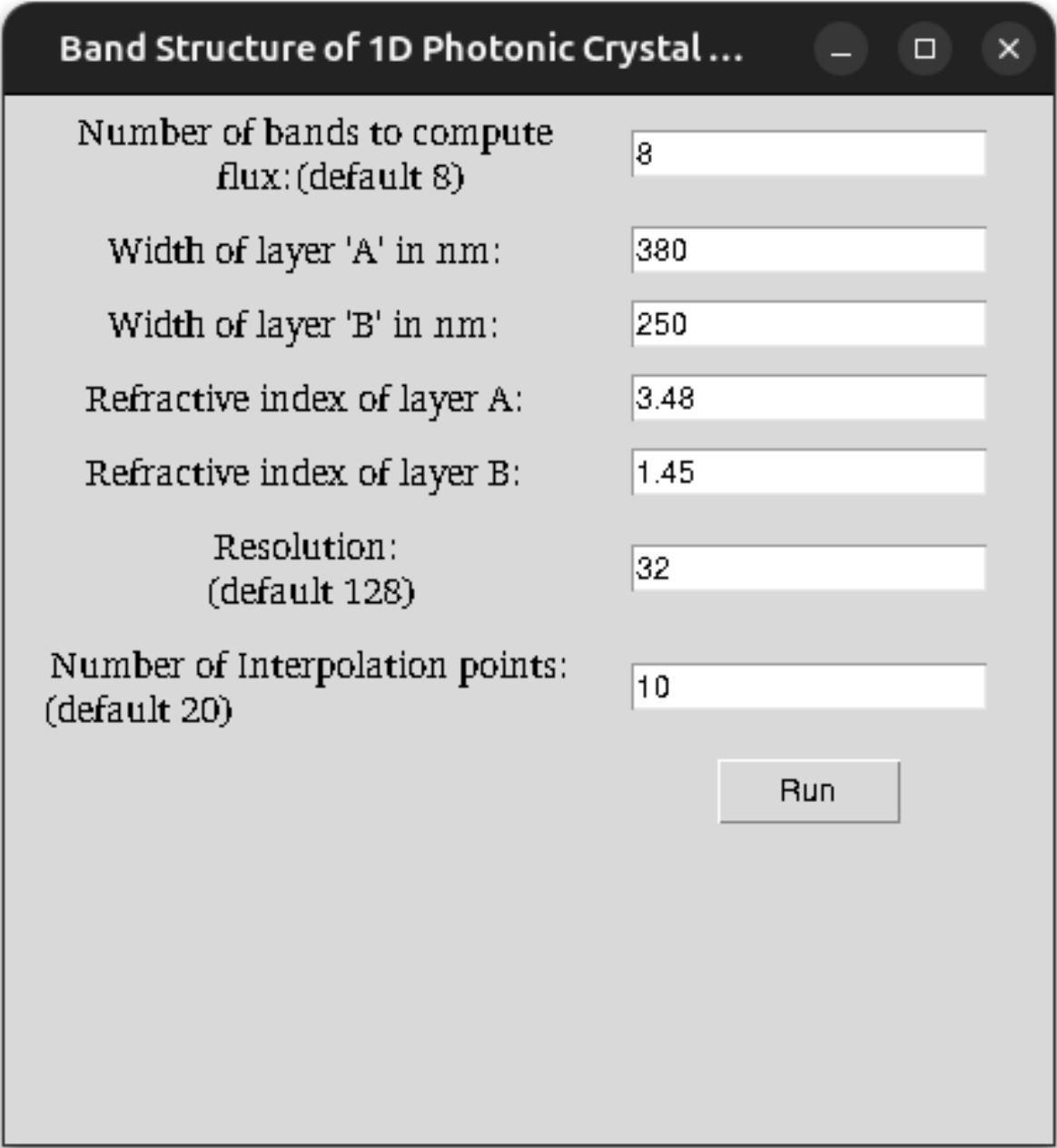


This will open a folder-creation pop-up as shown below. If we do not create a folder, the run happens in the parent directory or the source directory

So it is better to create a folder, and the source directory remains untouched



Then the parameter pop-up opens up as shown below; either default values can be used or any arbitrary values. It may be noted that currently 'nm' structures are only optimized

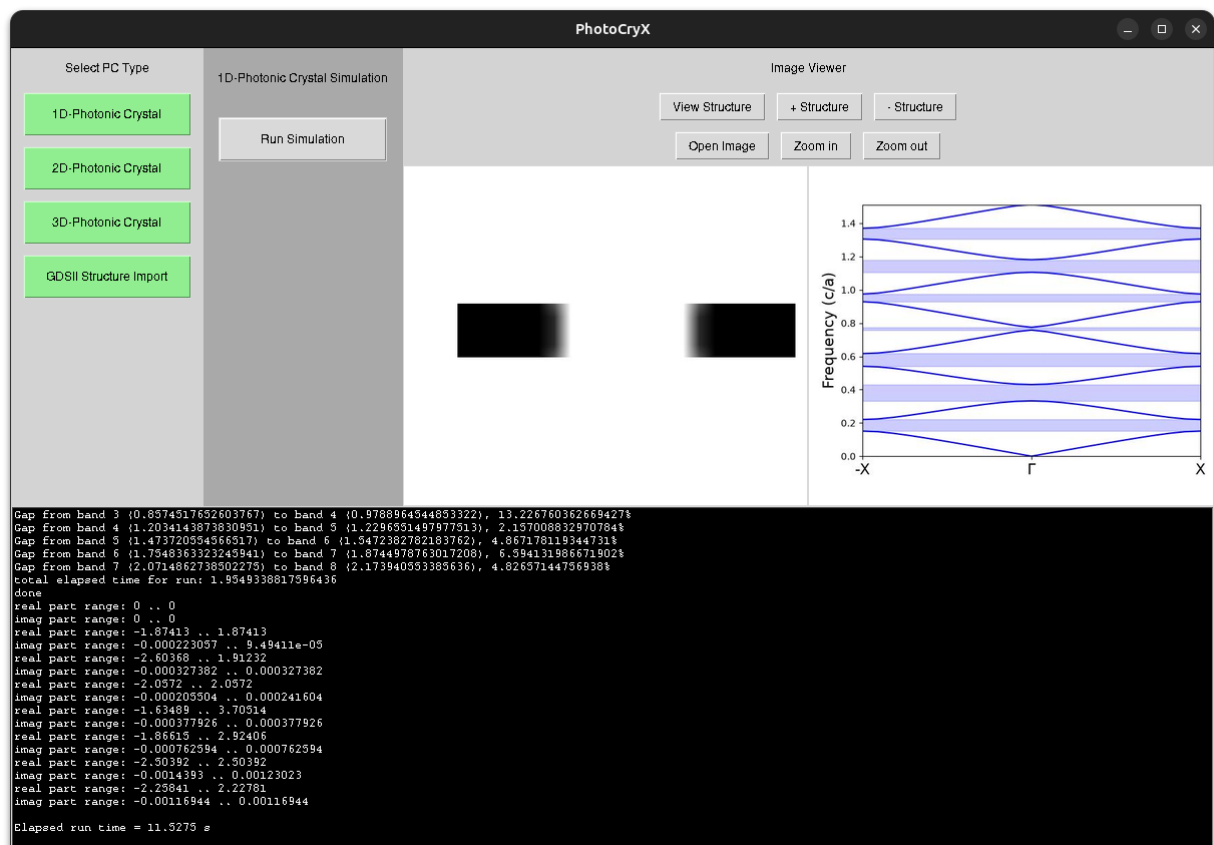


Band Structure of 1D Photonic Crystal ...

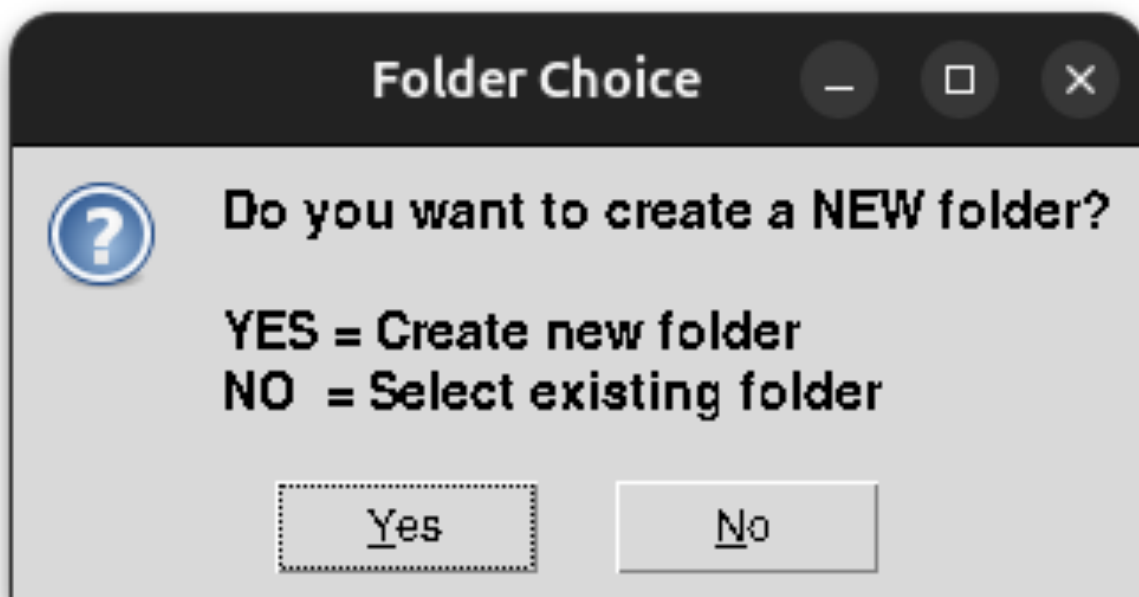
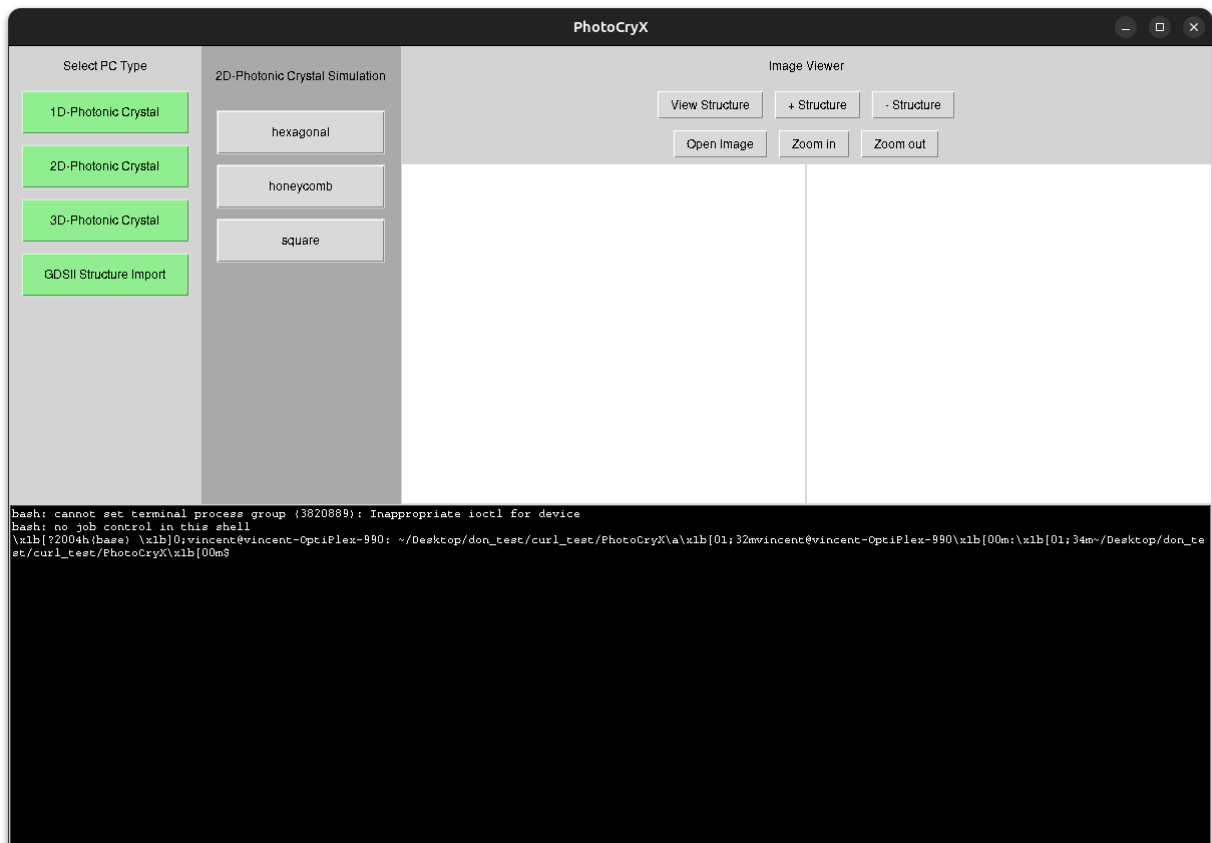
Number of bands to compute flux: (default 8)	8
Width of layer 'A' in nm:	380
Width of layer 'B' in nm:	250
Refractive index of layer A:	3.48
Refractive index of layer B:	1.45
Resolution: (default 128)	32
Number of Interpolation points: (default 20)	10

Run

The computed epsilon and band structure are displayed as shown; it also computes the field, which can be opened directly by clicking the "Open Image" tab



Two-dimensional Photonic Crystal (2DPC) - Honeycomb lattice



Band Structure of 2D Honeycomb Photoni...



Number of bands to compute
flux: (default 8)

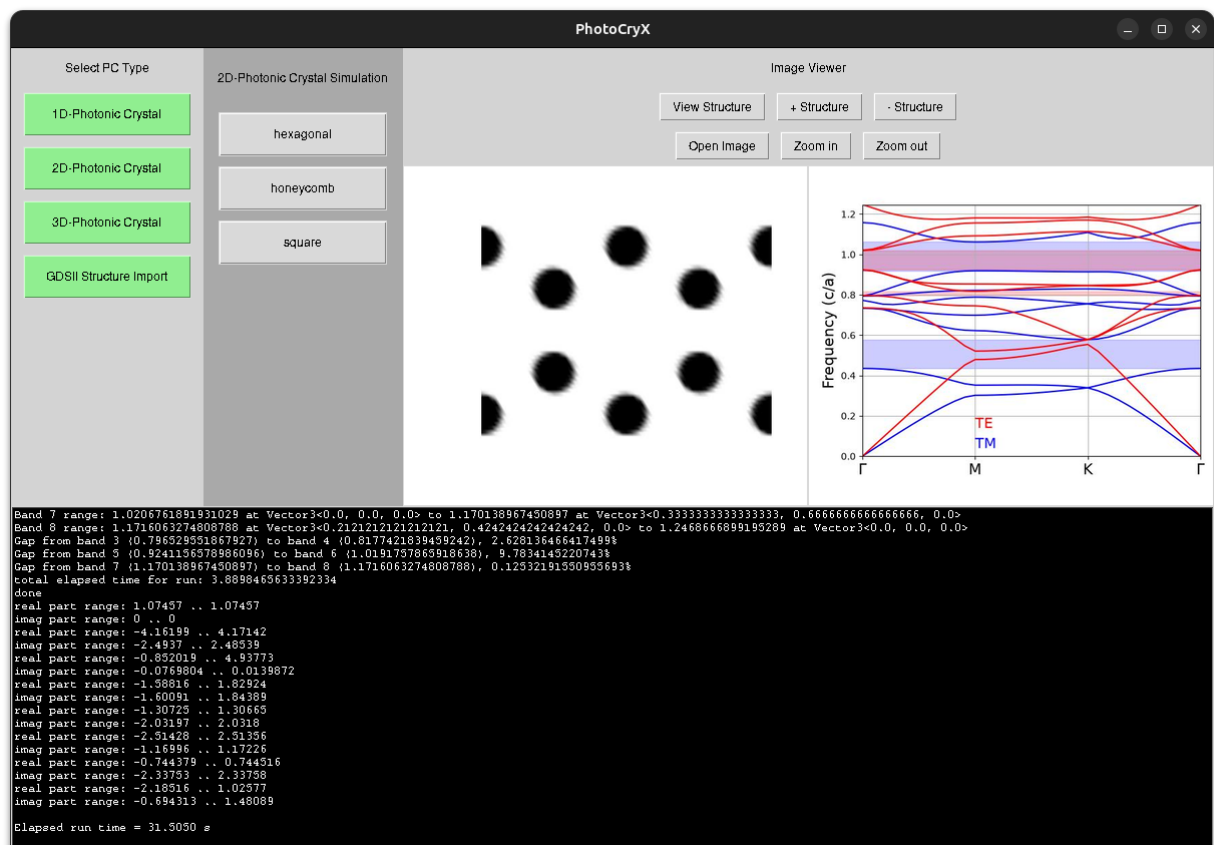
Side length 'a' in nm:
(default 1000)

r/a ratio:
(default 0.14)

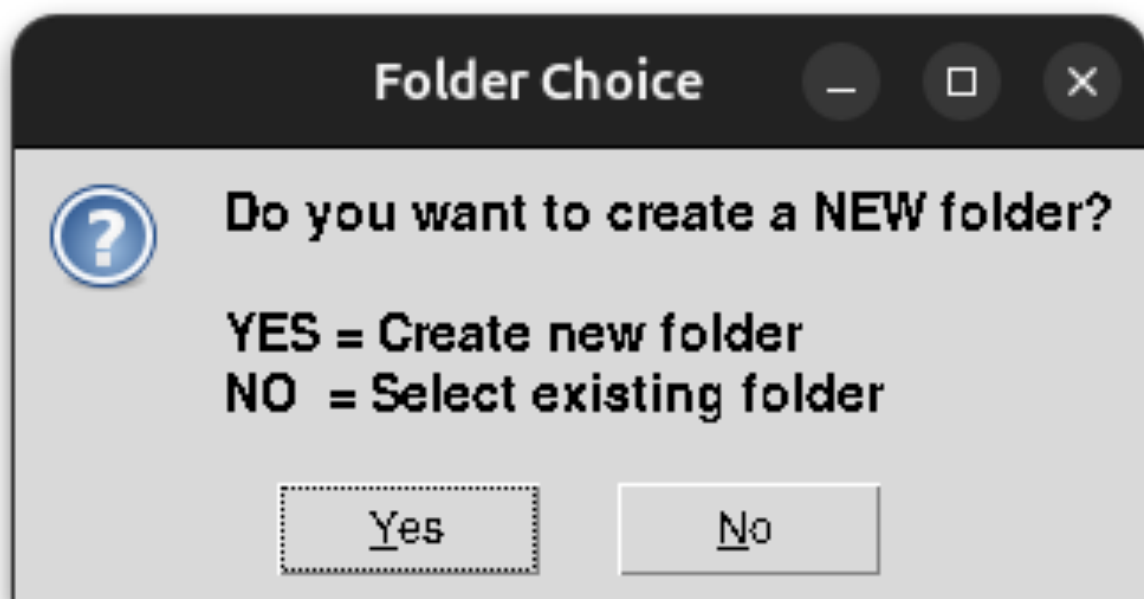
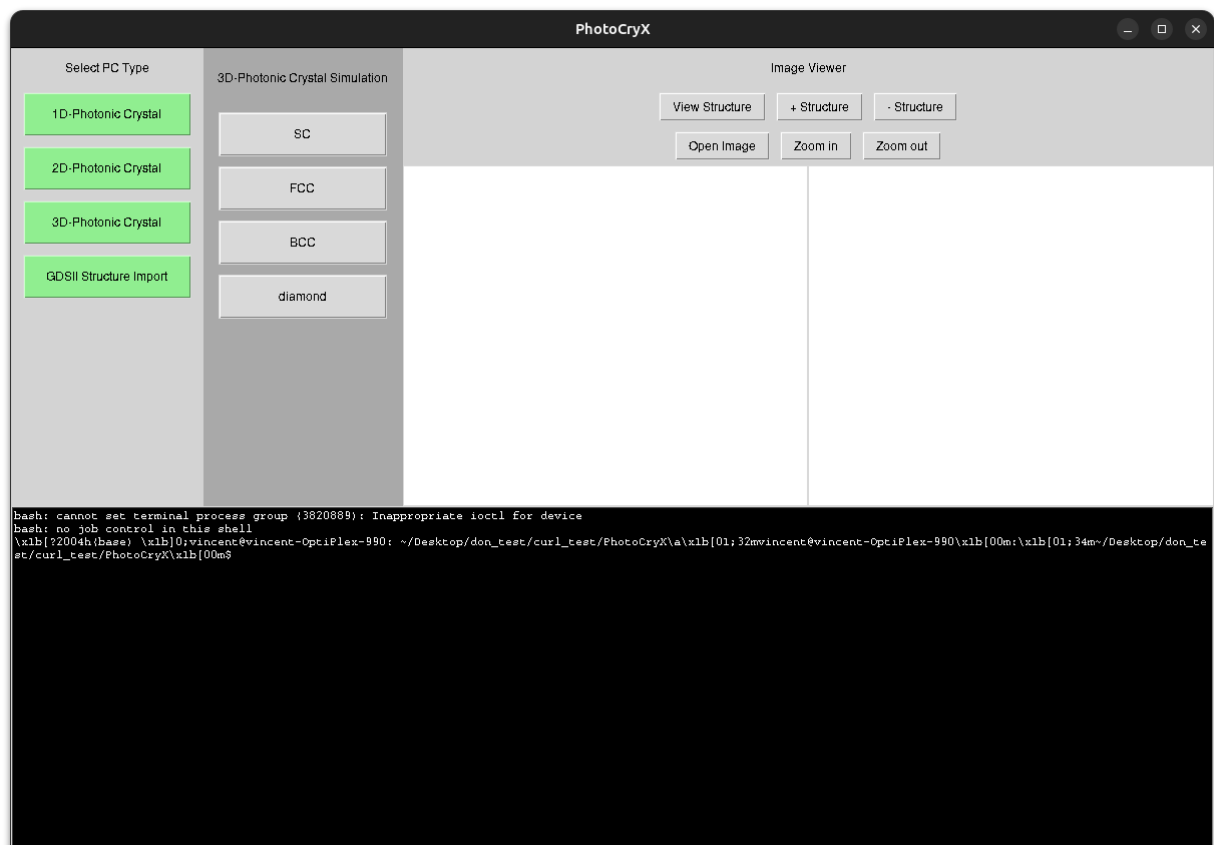
Resolution:
(default 32)

Dielectric constant of dielectric cylinder
(default 11.7)

Number of Interpolation points:
(default 20)



Three-dimensional Photonic Crystal (3DPC) - FCC lattice



Band Structure of 3D FCC Photonic Crysta...



Number of bands to compute
flux: (default 5)

Mesh Size:
(default 5)

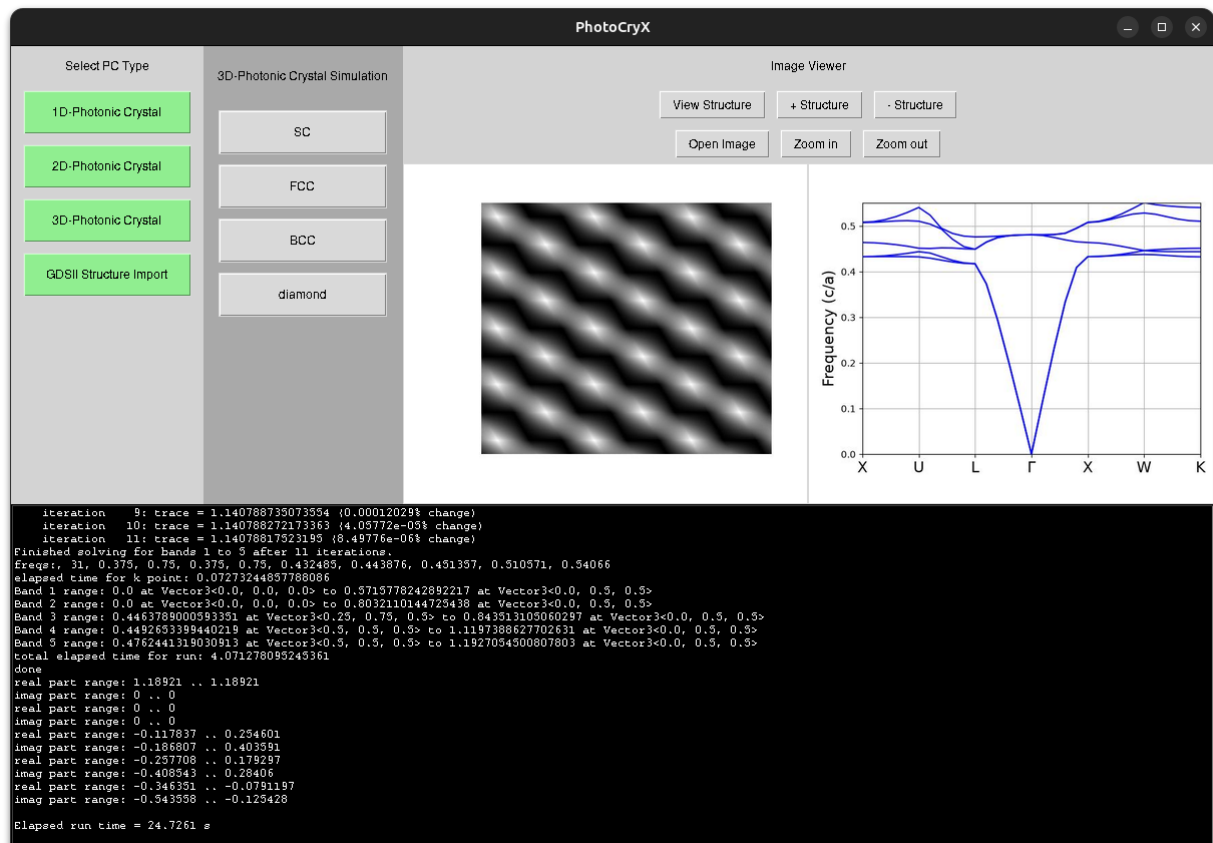
Resolution:
(default 16)

Dielectric constant of dielectric sphere
(default 13)

Radius of Sphere:
(default 0.3)

Number of Interpolation points:
(default 4)

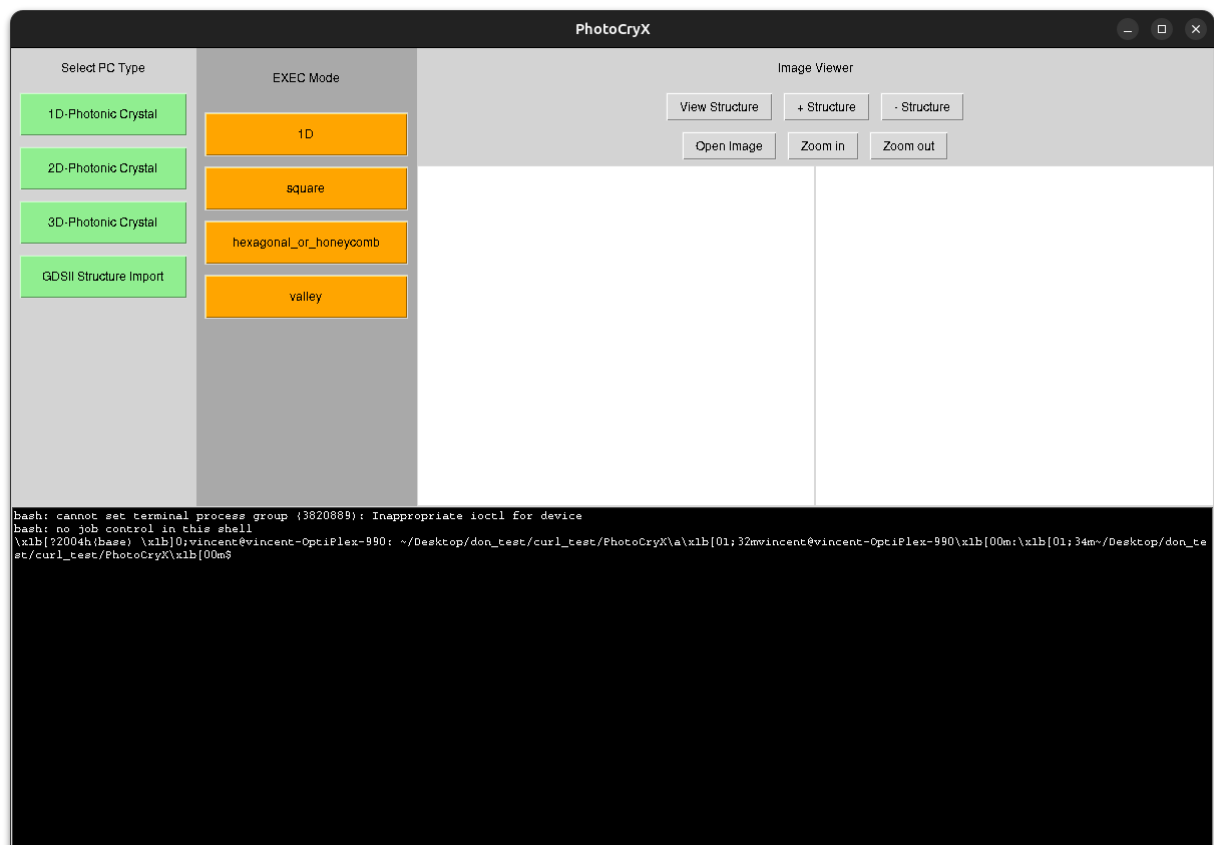
Run



The epsilon is just a 2D surface plot, to visualize in 3D we need additional packages like Mayavi or Vis5D, the installation is discussed [here](#)

All 3D folders are occupied with "plot3d.py" which uses Mayavi, The detailed 3D visualization is given [here](#)

Executable mode (EXEC) with GDSII file and "mpb.in" in the destination folder



The control file or the main input file is "mpb.in", here, a sample is given

```
# ===== Geometry Controls =====
GDS_FILE = dumbell.gds
INDEX = 11.7

# ===== Simulation Parameters =====
NUM_BANDS = 9
RESOLUTION = 128
INTERPOLATION_POINTS = 20

# ===== Plot Controls =====
PLOT_EPSILON = .TRUE.
PERIODS = 3
PLOT_BANDS = .TRUE.
PLOT_BANDS_NORM = .TRUE.

# KPATH = 0 2 1 0
#  $\epsilon$  = 0
# K = 1
# M = 2
# X = 4

KPATH = 2 0 1 2

FIELD_KPOINT = 0

PLOT_IM_FIELDS = .TRUE.
PLOT_IE_FIELDS = .TRUE.

PLOT_IM_FIELDS_NORM = .TRUE.
PLOT_IE_FIELDS_NORM = .TRUE.

PLOT_IM_PHASE = .TRUE.
PLOT_IE_PHASE = .TRUE.

#####
```